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Bridging Species Richness and True Diversity: A Quantitative and Ratio Analysis Approach from Indian Biodiversity Studies

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Authors' contributions

This work was carried out in collaboration among all authors. Author ADP conceptualized the study and performed the statistical analyses. Authors ADP, RD and SKA designed the study, developed the methodology, and wrote the first draft of the manuscript. All authors read and approved the final manuscript.

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ABSTRACT

Aims: The present study proposes new metrics linking species richness (N0) and true diversity, calculated as the first-order Hill Number (N1), viz., Diversity Delta (N0−N1), True Diversity-to-Species Richness Ratio (N1/N0), and Species Richness-to-True Diversity Multiple (N0/N1).

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Study Design: The proposed metrics were applied on a dataset of 351 observations from 105 published studies.
Place and Duration of Study: The analysis was based on observations drawn from ecological studies done across diverse ecosystems in India, published since 2000.
Methodology: Calculation of proposed measures and data visualization were performed using MS-Excel and Python. Incorrect and ambiguous observation were excluded from the dataset. Relationships between Shannon's Index (H) and species richness were also assessed.
Results: Diversity Delta ranged from 1 to 517, N0/N1 multiple from 1.0333 to 40.6667, and N1/N0 ratio from 0.0246 to 0.9677. Shannon's Index (H) exhibited a characteristic rectangular hyperbolic relationship with species richness (N0), highlighting diminishing returns of increasing richness on diversity.
Conclusion: The proposed metrics offer intuitive and complementary tools to conventional indices. While Diversity Delta (N0–N1) emerged as a robust sanity check to identify miscalculations, high N0/N1 multiples or low N1/N0 ratios acted as benchmarks to assess conservation success. The proposed ratios viz., N1/N0 Ratio and N0/N1 multiple offer insights analogous to financial ratios such as Earnings Per Share and Price-to-Earnings multiple, respectively, emphasizing per capita species contribution to true diversity rather than overall site-level equitability or dominance, providing a unique perspective distinct from existing measures. Additionally, the study highlights methodological ambiguities in reported diversity measures and their interpretation in publications, and proposes remedial measures to avoid those in future. The study also provides an exhaustive account of ecological work done in India over the last 25 years, documented in the references, serving as a valuable reference point for future investigations.

Keywords: Hill numbers; ratio analysis; species richness; true diversity.

1. INTRODUCTION

Biodiversity plays a critical role in sustaining ecosystems and supporting human survival and growth. Efforts to quantify biodiversity have led to the development of various metrics, including species richness, Simpson's Index, and Shannon's Index. Species richness, the simplest measure, quantifies the number of observed species. However, it lacks information on species abundances, leading to the development of diversity indices that integrate abundance with richness.

Simpson's Index, proposed by Simpson (1949), measures the probability of two randomly selected individuals belonging to the same species, mathematically expressed as:

Simpson's Index (D) = $\sum_{i=1}^{S} p_i^2 = \sum_{i=1}^{S} (n_i / N)^2$

Where, p_i (= n_i/N) is proportional abundance of ith species, n_i = number of individuals of ith species, N = total number of individuals (or isolates in case of microfungi) of all species, S= number of distinct species. Thus lower the value of the index, lower the probability of two individuals belonging to the same species, thus higher the diversity, and vice-versa. Its complement (1−D), termed Gini-Simpson Index (Jost, 2006), offers intuitive interpretation as higher values indicate greater diversity.

Shannon's index (Shannon & Weaver, 1949) measures the uncertainty in correctly predicting the species of a next randomly chosen individual, calculated as:

$$H = \sum_{i=1}^{S} p_i \ln(1 / p_i)$$

Where ln = natural logarithm.

Higher H values signify greater diversity. While these indices integrate richness and abundance, their units differ from species richness. This disconnect arises from their non-linearity, which stems from their being basically measures of probability and uncertainty, making them not directly comparable to species richness. One immediate consequence of this non-linearity is that unlike species richness, the diversity indices do not obey the doubling property. As a result, if two assemblages have an identical relative abundance distribution but share no species in common, on combining the two assemblages, species richness doubles, while diversity indices do not (Gotelli & Ellison, 2004). Hill Numbers provide a unifying framework, transforming diversity indices into "effective numbers of

species," which are comparable to species richness. Effective number of species, derived from a diversity index, is number of equally common taxa in a hypothetical assemblage for the given value of that diversity index. For example, if observed species richness is 20, but effective number of species is 7, true diversity is equivalent to that of a hypothetical assemblage consisting of 7 equally abundant species. The idea of true diversity and Hill Numbers was developed by MacArthur & Wilson (1967) and Hill (1970), and popularised by Jost (2006). Hill Numbers (q^D) are defined as:

$$q^D = \left(\sum_{i=1}^{s} p_i^q\right)^{1/(1-q)}$$

where q is an integer parameter determining 'order' of the particular Hill Number, which in turn determines the sensitivity of the calculated measure to common or rare species. As the order (q) increases, the common species get increasingly more weight, while opposite is true for rare species. q=0 gives species richness (N0); q=1 yields First Order Hill Number or Shannon's effective number of species ($N1=e^H$, where 'e' is Euler's number or natural base); and q=2 corresponds to Second Order Hill Number or Simpson's effective number of species (N2=1/D). Details on derivations can be found in Gotelli and Ellison (2004). In practice, both N1 and N2 are rounded down to nearest integer.

It is important to convert a diversity index into effective number of species (ENS_H) in order to make meaningful comparisons among different zones with regards to the diversity, since direct comparison of a diversity index is misleading due to its highly nonlinear nature (Jost 2006). For example, assume that first zone has H_1 = 4, and second zone has H_2 = 5. Direct comparison, solely on the value of *H*, will lead us to conclude that latter is 25% *more diverse than former*. However, by converting to true diversity, it can be seen that effective number of species for first zone (e^4) and second zone (e^5) are 54 and 148, respectively. Thus, difference in *true* diversity is of 174.07%, and not 25%.

Another reason for converting a diversity index into effective number of species is that doing so deflates species richness by accounting for abundance, thereby revealing *true* diversity. For illustration, consider a hypothetical forest with 1000 trees, out of which 900 trees belong to one species, rest of the 100 trees are of 100 different species. Here is the situation of a single dominant species accounting for 90% of total, whereas each of the 100 other species contributing 0.1% of the total. Here, Species Richness or Zeroth Order Hill Number = 101; Shannon's effective number of species or First Order Hill Number (^1D) $= e^{\Sigma p_i * \ln(1/p_i)} = e^{\{90\% * \ln(1/90\%) + 100 * [0.1\% * \ln(1/0.1\%)]\}} \approx 2$; Simpson's effective number of species or Second Order Hill Number (^2D) = $1/\ \Sigma p_i^2 = 1/[(90\%)^2 + 100*(0.1\%)^2] \approx 1$. The two hypothetical examples (Dubey & Pandey 2022a) above illustrate how species richness and diversity indices can obscure real diversity, while the Hill Number transformation effectively reveals it. Further insights can be gained from above hypothetical examples. First, the following relationship holds: N0 (Species richness) ≥ N1 ≥ N2. Second, species richness being zeroth order Hill Number overweighs rare species, while Simpson's index based Effective number of species being second order (q = 2) Hill Number overweighs common species (Gotelli & Ellison, 2004). It is for this reason that N1 has been more commonly used (e.g. Dubey & Pandey, 2022a,b,c; 2023) as it equally weighs both common and rare species. However, both measures (N0 & N1) seen in isolation project somewhat an incomplete picture.

In light of these, there arises a need for a unifying framework that offers simple, intuitive and easily interpretable measures, bridging species richness and true diversity. Therefore, we propose a novel framework based on difference-based metrics and ratio analysis—an approach widely used in finance—to better capture and interpret underlying biodiversity patterns and dominance effects across communities. The following section defines the proposed metrics and applies them to dataset from Indian biodiversity studies.

2. METHODOLOGY

2.1 Ratio Analysis: A Novel Framework for Biodiversity

Ratio analysis, widely applied in financial contexts, involves examining relationships between variables to extract meaningful insights, since absolute values of different variables on their own make little sense unless seen in relation to another (Reilly & Brown, 2012). Popular metrics like Earnings Per Share (EPS) and Price-to-Earnings (P/E) ratios offer interpretable insights into financial performance. Extending this framework, we propose a simple arithmetic measure and two biodiversity ratios to

bridge species richness (N0) and true diversity (N1):

Diversity Delta (N0−N1): It is the difference between species richness and true diversity. Larger values indicate greater dominance by a few species. In case of perfect equitability, when all species are equally abundant, the Diversity Delta will be equal to zero, since N0 ≈ N1. On the contrary, in a situation approaching monoculture (one species exercising overwhelming dominance), the Diversity Delta will be closer to N0, since N1 ≈ 1. As N0 ≥ N1, this also serves as a useful sanity check measure since a negative Diversity Delta is a consequence of miscalculation.

True Diversity/Species Richness Ratio (N1 /N0): This ratio quantifies the per capita contribution of species to diversity. Thus, higher the value, higher the diversity as lower the effect of dominant taxa, and vice versa. In other words, a lower value is due to lower average contribution of each species to the total diversity, as few dominant species contribute most of the diversity; on the other hand, perfect ratio (1 or 100%) means all species make equal average contribution to the diversity. Thus, higher values signify equitable communities with minimal dominance effects. The ratio is bounded between 0 and 1 (i.e. 0 < N1/N0 ratio ≤1). The expression is analogous to Earnings Per Share or EPS (Investopedia 2025a), used in financial analysis, which is net profit divided by number of outstanding common shares. Other things equal, a higher EPS is desirable as it reflects greater profitability—similarly, a higher N1/N0 implies a more balanced and diverse ecological community.

Species Richness/True Diversity Multiple (N0/N1): This measure highlights the influence of dominant species on community structure. In case of perfect equitability, where N0 ≈ N1, this multiple will be close to 1. On the contrary, in a monoculture like situation, the multiple will be close to N0, since N1 ≈ 1. Lower multiples, thus, indicate balanced communities with high effective diversity and lower dominance effects. The expression is analogous to Price to Earnings ratio or P/E multiple (Investopedia 2025b), used by financial analysts, which is market price of a share divided by earning per share. Other things equal, a high P/E ratio suggests overvaluation, just as a high N0/N1 multiple indicates lower community equitability and stronger dominance effects.

While the existing indices measure overall site-level equitability or dominance, the proposed metrics differ in emphasizing per capita species contribution to true diversity. These metrics, grounded in ratio analysis, thus complement traditional measures by offering intuitive interpretations of community structure and dominance effects.

Justification for choosing N1 Over N2: As N1 equally weighs both common and rare species, it is ideal for ratio analysis. For example, in aforementioned hypothetical forest with 101 species, where 90% of individuals belong to one species, N1= 2 and N2=1. The large Diversity Delta (=99), a high N0/N1 multiple (=50.5) and very low per species contribution to true diversity i.e. N1/N0 ratio (≈0.02 or 2%) signify a community characterized by low diversity despite relatively high species richness, owing to presence of a single dominant species. Using N2 in proposed ratios would exaggerate dominance effects (N0/N2 multiple will be 101, double of N0/N1 multiple), undermining their utility for ecological interpretation.

2.2 Data Collection and Analysis

Publications sourced from Google Scholar, ResearchGate, and Academia.edu were utilized to obtain ecological data. The dataset consists of 351 observations from 105 studies across India, covering diverse ecosystems, since 2000, mapped with QGIS 3.14 'Pi' version and also plotted on Google Map. Of these, 35 observations from 13 studies pertain to fungi, while the rest pertain to higher plants. Observations and associated calculated measures, unless stated otherwise, refer to the whole plant community. Observations yielding negative Diversity Delta values were excluded. Observations from studies lacking clarity on functional form of Shannon's Index by absence of explicit formulation or external reference from where it can be verified were also excluded. This is because Shannon & Weaver (1949) mention calculation of the index with different bases, and in ecological literature, Shannon's Index has been calculated with different bases, the most popular being natural base and base 2. Therefore, where Shannon's Index was calculated with base 2, we converted the value to natural base by multiplying with ln(2), the mathematical justification for which is given in supplementary information. Calculations were performed using MS-Excel and Python, with scripts and the full dataset (Table S1) included in

the supplementary material and are publicly available at Figshare: https://doi.org/10.6084/m9.figshare.28890035.

The dataset comprises observations from 105 representative ecological studies conducted across various regions of India. In the north, representative studies from Himachal Pradesh include Bisht et al. (2022) and Verma & Kapoor (2013). Jammu & Kashmir is represented by Shameem et al. (2010), and Ladakh by Joshi et al. (2006). Uttar Pradesh is represented by Gangwar & Lavania (2018) and Bajpai et al. (2020). Uttarakhand is represented by Gautam et al. (2006), De (2007), Chauhan et al. (2010), Devlal & Sharma (2010), Malik et al. (2014), Rawat et al. (2015), Akash et al. (2019), and Maletha et al. (2022).

In central India, representative studies from Madhya Pradesh include Patra et al. (2007), Thakur et al. (2008), Dar et al. (2019), Singh et al. (2021), and Jain et al. (2022). Chhattisgarh is represented by Kushwah & Kumar (2000) and Lal & Patil (2024).

From western India, Rajasthan is represented by Yadav & Gupta (2006), Sultana et al. (2014), and Mukherjee et al. (2017). Gujarat is represented by Joshi et al. (2011), Singh (2020), Chaudhari & Pathak (2022), Jha et al. (2023), Makvana & Paliwal (2023), and Sharma et al. (2024). Maharashtra is represented by Kunhikannan & Rao (2013), Datar (2016), Kunhikannan et al. (2017), Sahoo et al. (2017), Tadwalkar et al. (2020), Padmakumar & Murugan (2022), Dudipala et al. (2023), and Tamhane et al. (2024). Additional data for the state on microfungal ecology are from Dubey & Pandey (2022a–d; 2023). Goa is represented by Radhika & Rodrigues (2010) and D'Souza et al. (2013).

In southern India, representative studies from Andhra Pradesh include Reddy et al. (2008), Babu et al. (2010), Meerabai & Padmavathi (2011), Rao et al. (2011), Mastan & Reddy (2023), and Naidu et al. (2023). Telangana is represented by Murthy (2015), Suthari & Raju (2018), and Kumaraswamy et al. (2023). Karnataka is represented by multiple studies such as Nagaraja et al. (2005), Bhat et al. (2009), Chandran et al. (2010), Banakar et al. (2012), Sathish et al. (2013). Additional contributions are from Hegde et al. (2018), Neikha & Nagaraja (2019), Habib et al. (2020), Remadevi et al. (2020), and Revathy et al. (2023). Kerala is represented by Sharma et al. (2002), Menon & Suraj (2009), Juliya et al. (2012), Sundarapandian & Karoor (2013), Sreejith et al. (2016), Bilyaminu et al. (2021), and Akshaya et al. (2023). Tamil Nadu is represented by Pragasan & Parthasarathy (2010), Sundar et al. (2011), Thamizhmani & Senthilkumaran (2012), Gandhi et al. (2014), Anbarasu et al. (2015), Sundarapandian et al. (2015), Mohandass et al. (2016), Babu & Parthasarathy (2019), and Packialakshmi et al. (2022).

In eastern India, representative studies from Bihar and Jharkhand include Chandra et al. (2021) and Lal et al. (2019), respectively. West Bengal is represented by Basu et al. (2010), Rawat et al. (2018), Sinha et al. (2018), and Chettri et al. (2023). Odisha is represented by Reddy et al. (2007); Bhadra et al. (2010); Sahu et al. (2012); Panda et al. (2013); and Saravanan et al. (2019).

In the northeast, Sikkim is represented by Pandey et al. (2018) and Lahiri & Dash (2021); Assam by Sharma & Jha (2014) and Sharma et al. (2023). Arunachal Pradesh is represented by Deb et al. (2009), Bharali et al. (2011), and Saikia et al. (2017); Nagaland by Leishangthem & Singh (2018) and Ao et al. (2021). Meghalaya is represented by Tynsong et al. (2022) and Mizoram by Lalzarzovi & Lalnuntluanga (2022).

Among the islands, Andaman & Nicobar Islands are represented by Tripathi et al. (2004) and Ragavan et al. (2015), while Lakshadweep is represented by Reddy & Roy (2011).

3. RESULTS AND DISCUSSION

3.1 Results

The study documented a total of 351 observations from 105 studies across diverse geographies of India (Fig. 1). As the boxplots illustrate (Fig. 2), all variables except Shannon's Index and N1/N0 ratio exhibit substantial right skew (Appendix Fig. 1), with means greater than medians (Table 1) and several outliers indicating large observations among generally smaller values. Reported measures, unless stated otherwise, refer to the whole higher plant community.

Species Richness (N0): Values ranged from 2 to 549, with a standard deviation of 58.12. The interquartile range (IQR) – i.e. the range from 25th to 75th percentile, containing 50% of the observations – was 39, spanning from 15 to 54.

Mean (≈ 46.00) was exhibited by 3 locations viz., Eppanadu, Shola forest, Nilgiris, Tamil Nadu (Anbarasu et al., 2015); Doresanpalya, Near Bangalore, Karnataka (Remadevi et al., 2020); Saddle Peak forests [Inland], Andaman & Nicobar Islands (Tripathi et al., 2004). Median N0 (=26) was observed in locations like Great Himalayan National Park [disturbed site] (Bisht et al., 2022), Keoladeo National Park [Trees] (Mukherjee et al., 2017), and Kaziranga National Park [Grassland], Assam, Average for AM Fungi (Sharma & Jha, 2014) among many others. The minimum (=2) was observed in Namgia Valley [alt 4500-5000 m, for shrubs], Kinnaur, Himachal Pradesh (Verma & Kapoor, 2013), while the maximum (=549) was in Similipal Biosphere Reserve, Odisha (Reddy et al., 2007).

Shannon's Index (H): values ranged from 0.14 to 5.17, with a standard deviation of 0.96. The IQR (=1.19) spanned from 1.81 to 3.00, showing relatively less variability compared to other metrics. Mean Shannon's Index (≈ 2.42) exhibited by T.R. Pattinam, Tamil Nadu, for Mangrove Fungi (Thamizhmani & Senthilkumaran, 2012), closely followed by Nahargarh [H=2.43] and Ramgarh (H=2.44), Rajasthan (Sultana et al., 2014); Hardwickia forest (5/E4), Jhirpa [H=2.44], Madhya Pradesh (Singh et al., 2021). Median H (≈2.45) was observed, among others, in Tadoba National Park, for Savannah Trees (Kunhikannan & Rao, 2013) and Tapkeshwari, Kutch, Gujarat [Average] (Joshi et al., 2011). The minimum (≈0.14) and maximum (≈5.17) were observed in Nallamala, Andhra Pradesh, for Shrubs (Meerabai & Padmavathi, 2011), and Sanjay Gandhi National Park, Maharashtra, for microfungi (Dubey & Pandey, 2022a), respectively.

Shannon's Index (H) exhibited (Fig. 3) a characteristic rectangular hyperbolic relationship with species richness (N0). This relationship was modelled using Python's SciPy package, yielding an R^2 statistic of approximately 0.56, highlighting diminishing returns of increasing N0 on H.

True Diversity (N1): ranged from 1 to 176, with a standard deviation of 21.24. The IQR (=13.5, spanning from 6 to 19.5) indicates clustering of values in the lower range, but with some significant outliers. Mean N1 (≈ 17.00) exhibited by, among others, Tadoba National Park, Savannah, for Herbs (Kunhikannan & Rao 2013); Dry Deciduous Forest, Kawal Wildlife Sanctuary, Telangana (Murthy, 2015). Median N1 (=11) was recorded, among others, at Doon Valley, for herb layer (Chauhan et al., 2010) and Nagarhole Tiger Reserve [along highway stretch, average for trees], Karnataka (Habib et al., 2020). The minimum (=1) was observed at many locations, including Dudhwa National Park, for eucalyptus plantation (Bajpai et al., 2020); Kyongnosla Alpine Sanctuary, Sikkim, for trees [Average] (Lahiri & Dash, 2021). while the maximum (=176) was recorded in Sanjay Gandhi National Park, Maharashtra (Dubey & Pandey, 2022a).

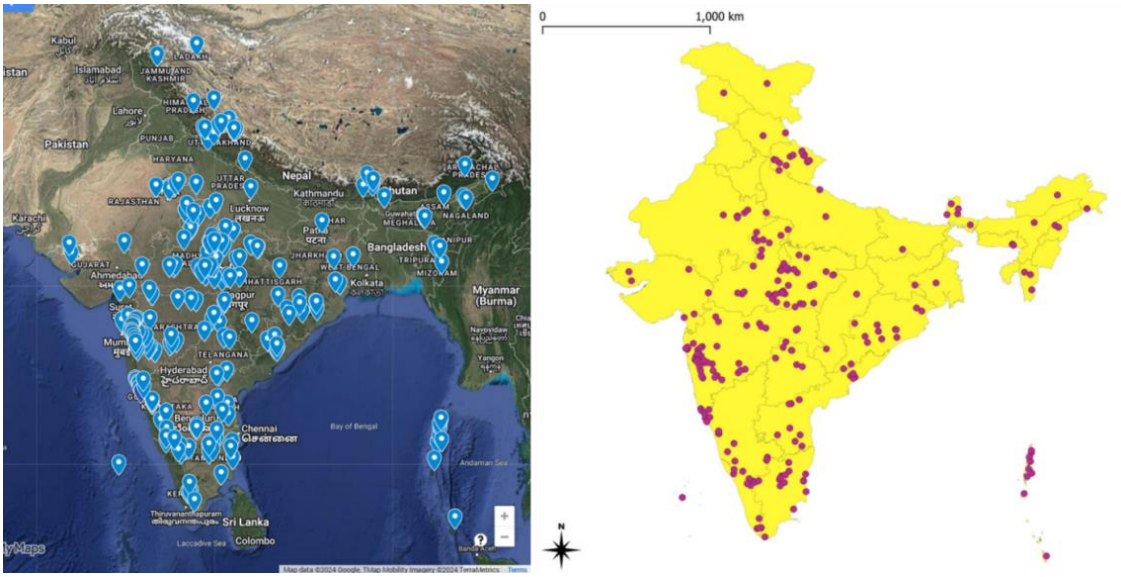

Fig. 1. Locations included in the dataset displayed on google maps and QGIS map (version 3.14 'pi')

Diversity Delta (N0−N1): This measure ranged from 1 to 517, with a standard deviation of 55.09. The IQR (=20) spanned from 7 to 27. Mean N0-N1 (≈ 29.00) exhibited by, among others, Gandhmardhan Hills, Odisha (Bhadra et al., 2010); Kuldiha Wildlife Sanctuary, Odisha [Core] (Saravanan et al., 2019); Dahuli, Maharashtra (Tamhane et al., 2024). Median N0-N1 (≈ 12.00) exhibited by, among others, Supe, Maharashtra [grassland] (Datar, 2016); Panna Tiger Reserve, Madhya Pradesh [Trees] (Jain et al., 2022); Subbaiahpalem, Andhra Pradesh [Thorn Scrub] (Naidu et al., 2023); Nanda Devi Biosphere Reserve, Lata-Toma-Phagti, Uttarakhand [Average for Shrubs] (Rawat et al., 2015); Navsari, Gujarat [for mangroves] (Singh, 2020). The minimum (=1.00) was observed at multiple sites, including Rudraprayag, Uttarakhand (Devlal & Sharma, 2010), while the maximum (=517) was seen in Similipal Biosphere Reserve, Odisha (Reddy et al., 2007).

The N1/N0 ratio ranged from 0.0246 to 0.9677, with IQR (=0.33) spanning from 0.28 to 0.61; while the N0/N1 multiple ranged from 1.03 to 40.67, with IQR (=1.89) spanning from 1.63 to 3.52. Mean N1/N0 ratio (≈ 0.45) was observed for Tadoba Tiger Reserve, Maharashtra [Average for Trees] (Kunhikannan et al., 2017), Periyar WLS (Sundarapandian & Karoor, 2013). Mean N0/N1 multiple (=4) was exhibited by, among others, Grassland, Supe (Datar, 2016); Lankai Area, Namdapha National Park, Arunachal Pradesh (Deb et al., 2009). The median N1/N0 ratio (≈ 0.46) and median N0/N1 multiple (≈ 2.17) were observed for Katghora Forest Division, Chhattisgarh (Lal & Patil, 2024) and Ziziphus scrub (6B/DS1), Pohri, Shivpuri, Madhya Pradesh (Singh et al., 2021), with additional median N1/N0 for Phawngpui National Park, Trees, Mizoram (Lalzarzovi & Lalnuntluanga, 2022). Due to their inverse mathematical relationship, the minimum N0/N1 multiple (≈1.03) and maximum N1/N0 ratio (≈0.97) were both observed in Pune city garden area, for foliar fungi (Dubey & Pandey, 2023). Conversely, the maximum N0/N1 multiple (≈40.67) and minimum N1/ N0 ratio (≈0.02) were recorded in Uttara Kannada Sacred Swamps (Hegde et al., 2018). Enumeration of all locations where ties occurred across various measures is provided in the Appendix.

3.2 Discussion

This study introduces three simple measures linking N0 to N1: Diversity Delta, N1/N0 ratio, and N0/N1 multiple. These metrics provide intuitive insights into community structure and species dominance. Diversity Delta quantifies the spread between N0 and N1, with a larger spread indicating greater dominance effects. The N1/N0 ratio describes the per capita contribution of species to true diversity, offering insights into the average role of a species. The N0/N1 multiple explicitly highlights the influence of dominant species in the ecosystem.

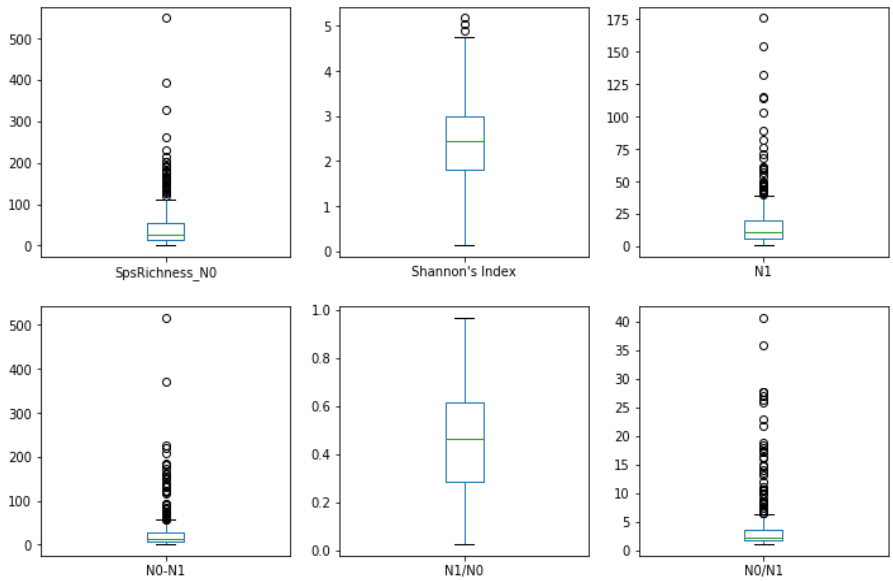

Fig. 2. Box-whisker plots of different diversity measures viz., Species richness (N0), Shannon's Index, True Diversity (N1), Diversity Delta (N0-N1), N1/N0 ratio, N0/N1 multiple

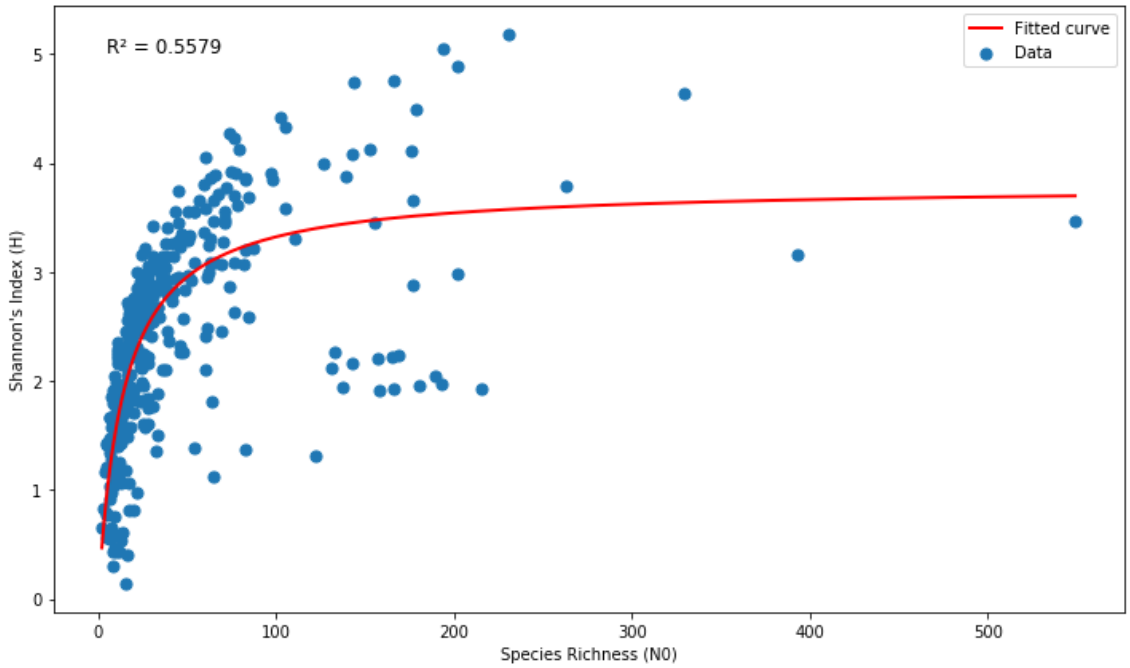

Fig. 3. Relationship between species richness (N0) and Shannon's Index (H)

Table 1. Descriptive Statistics of various diversity measures

	Species Richness (N0)	Shannon's Index (H)	N1	Diversity Delta (N0-N1)	N1/N0 Ratio	N0/N1 Multiple
count	351	351	351	351	351	351
mean	45.9316	2.4204	17.2051	28.7265	0.4501	4.0249
Standard deviation	58.1150	0.9450	21.2385	49.8560	0.2283	5.3283
minimum	2	0.1427	1	1	0.0246	1.0333
25%	15	1.8100	6	7	0.2839	1.6296
50%	26	2.4500	11	12	0.4615	2.1667
75%	54	2.9964	19.5	27	0.6137	3.5227
maximum	549	5.1740	176	517	0.9677	40.6667

We applied these measures to 351 observations from 105 published studies across India, representing diverse geographical regions (Fig. 1). The dataset spans ecosystems ranging from tropical forests, islands and grasslands to alpine zones, providing a comprehensive view of India's ecological diversity. The present study also provides a comprehensive account of ecological studies from different parts of India published since 2000.

Most diversity metrics, except Shannon's Index and N1/N0 ratio, exhibited substantial right skew (Fig. 2). This reflects the ecological heterogeneity of India, where values for species richness (N0) range from 2 in Namgia Valley, Kinnaur, Himachal Pradesh (Verma & Kapoor, 2013), to 549 in Similipal Biosphere Reserve, Odisha (Reddy et al., 2007). True diversity (N1) values range from 1 to 176, with locations like Sanjay Gandhi National Park recording the highest N1 for microfungi (Dubey & Pandey, 2022a). Outliers and large observations of species richness (N0), as also corresponding values of N1, deal with calculations done for larger geographical areas by pooling up data of constituent smaller subregions.

The proposed metrics reveal significant ecological patterns. Diversity Delta (N0−N1) identifies the dominance effect in ecosystems. For example, Similipal Biosphere Reserve (Reddy et al., 2007) exhibited the highest Diversity Delta (N0−N1=517), reflecting high richness with substantial dominance, likely due to pooled regional data. The N1/N0 ratio offers a

direct measure of per capita species contribution and its inverse N1/N0 offers insight into the dominance effects. For example, Pune city garden foliar fungi (Dubey & Pandey, 2023) exhibits near-perfect equitability (N1/N0=0.97), while Uttara Kannada Sacred Swamps (Hegde et al., 2018) show extreme dominance by a few species with low N1/N0 ratio (=0.02) and high N0/N1 multiple (=40.67).

Shannon's Index (H) exhibits a characteristic rectangular hyperbolic relationship with species richness (N0), as evident from the dataset. This relationship, modelled with Python's SciPy package, yielded an R^2 statistic of approximately 0.56. The diminishing returns of increasing N0 on H highlight the non-linear influence of species richness on diversity. This behaviour aligns with the theoretical understanding of how evenness and richness contribute differently to overall diversity measures.

Among the three metrics, Diversity Delta (N0−N1) emerged as a valuable sanity check. Since N1 must deflate N0 (and not inflate it), a negative Diversity Delta indicates computational errors. This screening helped exclude observations that would otherwise compromise the analysis. This sanity check measure will also be extremely helpful for readers, reviewers and decision makers in ascertaining the relevance of the research findings. From a conservation perspective, the proposed measures highlight the importance of increasing N1 (effective species richness) through habitat management, reforestation, or reducing dominance by a few species. For instance, decreasing the N0/N1 multiple or increasing the N1/N0 ratio can serve as benchmarks for monitoring the success of conservation efforts over time. Their simplicity and interpretability make these metrics accessible to ecologists, policymakers, and conservationists alike.

The proposed metrics offer a significant departure from traditional indices that focus on site-level equitability or dominance. By emphasizing per capita species contributions to true diversity, these ratio-based metrics provide intuitive insights into community structure and dominance effects, making them a valuable complement to established measures. This nuanced perspective enhances our understanding of biodiversity patterns and informs targeted conservation strategies.

The analysis also revealed methodological inconsistencies in published studies. Many excluded observations lacked clarity on the functional form of diversity indices. For example, Shannon's Index is historically calculated with both base 2 and the natural base. A value of 3.2 (base 2) corresponds to 2.2 (natural base), and the resultant N1 differs vastly (24 vs. 9), making comparisons unreliable without clear reporting. Older conventions of denoting natural base logarithms as "log" instead of "ln" further exacerbate this issue, as current convention is to use 'log' to denote base 10. Similarly, inconsistencies were noted in reporting Simpson's Index. Its complement (1−D) is often mislabelled as "Simpson's Index," though it should be termed "Gini-Simpson Index" (Jost, 2006). Some studies also misrepresent second-order Hill number (N2) as Simpson's Index. Such ambiguities persist in software packages, necessitating better standardization in reporting and interpretation. Current practices should align with standard nomenclature to minimize confusion.

Despite the breadth of the dataset, several ecological gaps remain. Numerous regions, including protected areas, lack sufficient biodiversity data. Recent studies across varied Indian ecosystems—including the Himalayas, Western Ghats, and central and western India—report wide-ranging patterns in plant species diversity, composition, and regeneration, often shaped by altitude, habitat type, seasonality, and anthropogenic pressures (Jishtu et al. 2025; Musisi et al. 2025; Pawar & Mule 2025; Reang et al. 2025; Sekar et al. 2025; Tank et al. 2025). Additionally, fungal diversity remains highly underexplored, with only 35 fungal observations across 13 studies in the current dataset. Yet fungi recorded the highest Shannon's Index and N1 values (Dubey & Pandey, 2022a), highlighting their immense but understudied diversity. With recent global fungal diversity estimates ranging from 2.2 to 3.8 million species (Hawksworth & Lücking, 2017), there is a clear need for targeted research on this critical group. Establishing representative permanent plots for periodic assessments across diverse geographies will enhance our understanding of biodiversity and ecological dynamics.

4. CONCLUSION

The proposed metrics—Diversity Delta, N1/N0 ratio, and N0/N1 multiple—offer novel yet intuitive approaches for interpreting biodiversity patterns and evaluating conservation outcomes. Combined with standardized reporting of

diversity indices and systematic monitoring, they have the potential to advance ecological research and biodiversity conservation efforts on both regional and global scales. Additionally, this study compiles a comprehensive record of ecological research conducted across India over the past 25 years. This collection serves as a valuable resource and reference point for future ecological investigations.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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APPENDIX

Converting Shannon's Index to desired base:

Another popular version of Shannon's Index with base 2 is given as:

$$\mathrm{H}_2 = -\sum_{i=1}^{S} p_i \log_2(p_i)$$

Rule of change of logarithm of base from a to b states that $\log_a(x) = \log_b(x) / \log_b(a)$

Therefore, to change the base from 2 to natural base, we have

$$-\sum_{i=1}^{S} p_i \log_2(p_i) = \frac{-\sum_{i=1}^{S} p_i \, ln(p_i)}{\ln(2)}$$

$H_2 = \frac{\mathrm{H}_e}{\ln(2)}$ i.e. $\mathrm{H}_e = \ln(2) * \mathrm{H}_2$

Detailed Results of Various Metrics:

Mean SpsRichness_N0 (≈ 46.00) exhibited by 3 locations viz., Eppanadu, Shola forest, Nilgiris, Tamil Nadu (Anbarasu et al. 2015); Doresanpalya, Near Bangalore, Karnataka (Remadevi et al. 2020); Saddle Peak forests [Inland], Andaman & Nicobar Islands (Tripathi et al. 2004). Median SpsRichness_N0 (≈ 26.00) exhibited by 10 locations viz., Great Himalayan National Park (GHNP) [Disturbed] (Bisht et al. 2022); Maharashtra evergreen forests Foliar fungi (Dubey & Pandey 2023); Keoladeo National Park, Rajasthan, Trees (Mukherjee et al. 2017); North Goa, AM Fungi (Radhika & Rodrigues 2010); Dachigam National Park, Jammu & Kashmir, Summer (Shameem et al. 2010); Kaziranga National Park, Grassland, for AM Fungi [Average] (Sharma & Jha 2014); Ramgarh, Rajasthan (Sultana et al. 2014); Bahadurpur and Ranipura, Sagar dist., Madhya Pradesh (Thakur et al. 2008); Muthukuda, Tamil Nadu, for mangrove fungi (Thamizhmani & Senthilkumaran 2012). Minimum SpsRichness_N0 (≈ 2.00) exhibited by 1 location viz., Namgia Valley [alt 4500-5000 m, for shrubs], Kinnaur, Himachal Pradesh (Verma & Kapoor 2013). Maximum N0 (≈ 549.00) was exhibited by 1 location viz., Similipal Biosphere Reserve, Odisha [Total] (Reddy et al. 2007)

Mean (≈ 2.42) and Median = 2.45) for Shannon's Index were very close. Mean Shannon's Index (≈ 2.42) exhibited by T.R. Pattinam, Tamil Nadu, for Mangrove Fungi (Thamizhmani & Senthilkumaran 2012), closely followed by Nahargarh [H=2.43] and Ramgarh (H=2.44), Rajasthan (Sultana et al. 2014); Hardwickia forest (5/E4), Jhirpa [H=2.44], Madhya Pradesh (Singh et al. 2021). Median Shannon's Index was exhibited by 4 locations viz., Nagarhole Tiger Reserve, along State Highway 33, Karnataka (Habib et al. 2020); Tapkeshwari [Average] (Joshi et al. 2011); Central Forest Circle, Kerala, for fungi (Juliya et al. 2012); Tadoba National Park, Savannah Trees, Maharashtra (Kunhikannan & Rao 2013). Minimum Shannon's Index (≈ 0.14) exhibited by 1 location viz., Nallamala, Andhra Pradesh, for shrubs (Meerabai & Padmavathi 2011). Maximum Shannon's Index (≈ 5.17) exhibited by 1 location viz., Sanjay Gandhi National Park (SGNP), Maharashtra, for Microfungi [Total] (Dubey & Pandey 2022a).

Mean N1 (≈ 17.00) exhibited by 8 locations viz., Tadoba National Park, Savannah, for Herbs (Kunhikannan & Rao 2013); Dry Deciduous Forest, Kawal Wildlife Sanctuary, Telangana (Murthy 2015); North and South Goa, for AM Fungi (Radhika & Rodrigues 2010); South Andaman (Ragavan et al. 2015); Khair forest (5/1S1), Pohri, Shivpuri and Dry teak forest (5A/C1B), North Betul, Madhya Pradesh (Singh et al. 2021); Mainpani, Sagar, Madhya Pradesh(Thakur et al. 2008).

Median N1 (≈ 11.00) was exhibited by 12 locations viz., Doon Valley, for Herb layer (Chauhan et al. 2010); Nagarhole Tiger Reserve, along State Highway 33 stretch, for Trees [Average] (Habib et al. 2020); Tapkeshwari, Gujarat [Average] (Joshi et al. 2011); Central Forest Circle, Kerala, for fungi (Juliya et al. 2012); Tadoba National Park, Savannah, Trees (Kunhikannan & Rao 2013); Hardwickia

forest (5/E4), Jhirpa (Singh et al. 2021); Ramgarh and Nahargarh, Rajasthan (Sultana et al. 2014); Periyar WLS, severely disturbed, for trees [Total] (Sundarapandian & Karoor 2013); Adilabad (Suthari & Raju 2018); T.R. Pattinam, Tamil Nadu, for Mangrove Fungi (Thamizhmani & Senthilkumaran 2012); Foothill forest, Saddle Peak, Andaman (Tripathi et al. 2004).

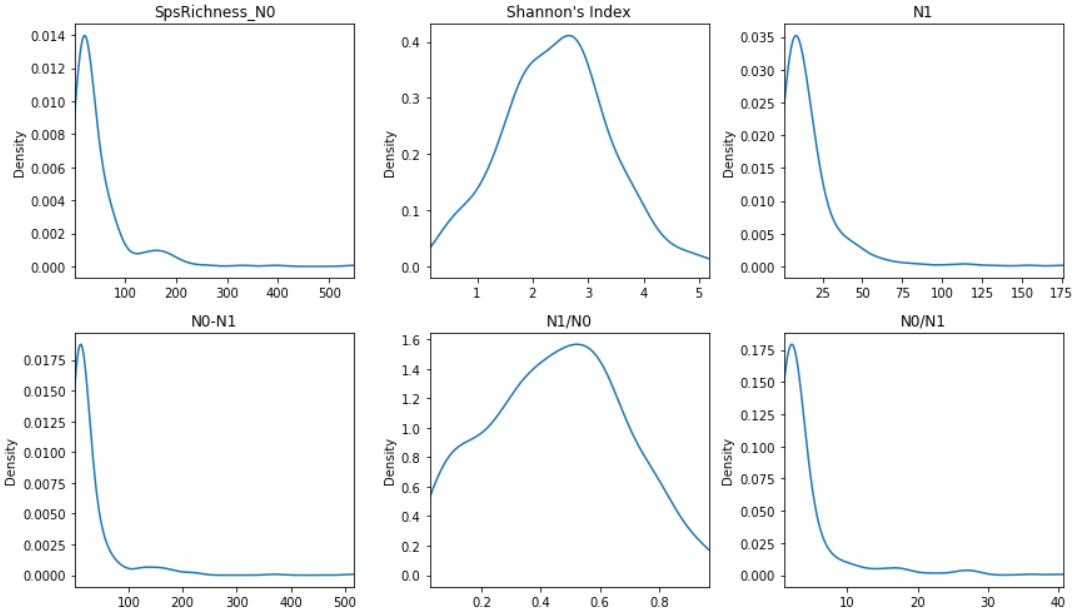
Minimum N1 (≈ 1.00) exhibited by 16 locations viz., Dudhwa National Park, eucalyptus plantation, Uttar Pradesh (Bajpai et al. 2020); Grasslands at Atpadi and Tikona, Maharashtra (Datar 2016); Mangrove AM Fungi of Zuari, Mandovi, Galgibagh, Sal, Terekhol, and Chapora, Goa (D'Souza et al. 2013); Kukrail (Minimum), Uttar Pradesh (Gangwar & Lavania 2018); Balai Territorial Forest, Madhya Pradesh, for trees (Jain et al. 2022); Nallamala, for trees and shrubs (Meerabai & Padmavathi 2011); Slightly moist teak forest (3B/C1C), Banapura, Hoshangabad, Madhya Pradesh (Singh et al. 2021); Namgia Valley [alt 4500-5000 m, for shrubs], Kinnaur (Verma & Kapoor 2013); Kyongnosla Alpine Sanctuary, Sikkim, for trees [Average] (Lahiri & Dash 2021). Maximum N1 (≈ 176.00) exhibited by 1 location viz., SGNP, for Microfungi (Dubey & Pandey 2022a)

Mean N0-N1 (≈ 29.00) exhibited by 4 locations viz., Nagarhole Tiger Reserve, along State Highway 33 stretch, Shrubs, Average (Habib et al. 2020); Gandhmardhan Hills, Odisha (Bhadra et al. 2010); Kuldiha Wildlife Sanctuary, Odisha [Core] (Saravanan et al. 2019); Dahuli, Maharashtra (Tamhane et al. 2024). Median N0-N1 (= 12.00) exhibited by 14 locations viz., Supe and Laling, Maharashtra [grasslands] (Datar 2016); Panna Tiger Reserve, Madhya Pradesh [Trees] (Jain et al. 2022); Subbaiahpalem, Vishakhapatnam, Andhra Pradesh [Thorn Scrub] (Naidu et al. 2023); Nanda Devi Biosphere Reserve, Lata-Toma-Phagti, Uttarakhand [Average for Shrubs] (Rawat et al. 2015); Navsari, Gujarat [for mangroves] (Singh 2020); Southern moist mixed deciduous forest (3B/C2), Alirajpur, Kathiwara (Singh et al. 2021); Jhalana, Rajasthan (Sultana et al. 2014); Periyar WLS [Moderately and Severely Distrurbed, for Herbs (Total)] (Sundarapandian & Karoor 2013); Ranipura, Sagar (Thakur et al. 2008); Muthukuda, for mangrove fungi (Thamizhmani & Senthilkumaran 2012); Namgia Valley, Kinnaur, alt 4500-5000 m, for Herbs (Verma & Kapoor 2013); Kyongnosla Alpine Sanctuary, Trees [Average] (Lahiri & Dash 2021). Minimum N0-N1 (≈ 1.00) exhibited by 12 locations viz., Rudraprayag, Uttarakhand, for trees [Minimum, Average and Maximum] (Devlal & Sharma 2010); Peripheral areas and Krishnagiri, SGNP, Maharashtra, for Microfungi (Dubey & Pandey 2022a); Pune City Garden area, for Foliar fungi (Dubey & Pandey 2023); Gorai Mangroves, Mumbai, Maharashtra (Padmakumar & Murugan 2022); Ravine thorn forest (6B/C2), Goraghat, Datia, and Dry grassland (5/DS4), Sheopur, Madhya Pradesh (Singh et al. 2021); Kanyakumari, Tamil Nadu, for VAM fungi [Average] (Sundar et al. 2011); Namgia Valley, alt 4000-4500 m and 4500-5000 m, Kinnaur, for Shrubs (Verma & Kapoor 2013). Maximum N0-N1 (≈ 517.00) exhibited by 1 location viz., Similipal Biosphere Reserve [Total] (Reddy et al. 2007)

Mean N1/N0 ratio (≈ 0.45) was observed for 7 locations: Tadoba Tiger Reserve, Maharashtra [Average for Trees] (Kunhikannan et al. 2017), Cuddalore, Palvathunnan, Tamil Nadu, for disturbed area (Babu & Parthasarathy 2019); Southern tropical dry deciduous forest, Udainagar, Dewas and Dry peninsular Sal forest, Gopharu, Shahdol, Madhya Pradesh (Singh et al. 2021); Periyar WLS, Kerala [Moderately Disturbed, for Shrubs; Severely Disturbed, for Herbs](Sundarapandian & Karoor 2013); Abhepuri, Maharashtra (Tamhane et al. 2024).

Mean N0/N1 multiple (≈ 4.00) was exhibited by 5 locations viz., Dudhwa National Park, teak plantation (Bajpai et al. 2020); Grassland, Supe, Maharashtra (Datar 2016); Lankai Area, Namdapha National Park (Deb et al. 2009); Hingana Territorial Forest, Madhya Pradesh, for Trees (Jain et al. 2022); Middle Saddle peak forest, Andaman & Nicobar Islands (Tripathi et al. 2004).

The median N1/N0 ratio (≈ 0.46) and median N0/N1 multiple (≈ 2.17) were observed for Katghora Forest Division, Chhattisgarh (Lal & Patil 2024) and Ziziphus scrub (6B/DS1), Pohri, Shivpuri, Madhya Pradesh (Singh et al. 2021), with additional median N1/N0 for Phawngpui National Park, Mizoram, for trees (Lalzarzovi & Lalnuntluanga 2022). Due to their inverse mathematical relationship, the minimum N0/N1 multiple (≈1.03) and maximum N1/N0 ratio (≈0.97) were both observed in Pune city garden area, foliar fungi (Dubey & Pandey, 2023). Conversely, the maximum N0/N1 multiple (≈40.67) and minimum N1/ N0 ratio (≈0.02) were recorded in Uttara Kannada Sacred Swamps, Karnataka (Hegde et al., 2018).



Appendix Fig. 1. Density plots of various diversity measures

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